CS24307 — THEORY (3.0 CR) MODULE V NOTES BIT MESRA | NEP SCHEME

Module 5: Machine Learning & Neural Networks

Decision Trees (ID3 Gain), Perceptron, Backpropagation & Bias-Variance

Module V: Machine Learning Foundations — Topics Covered

1. Inductive Learning & Decision Trees (ID3)
2. Entropy & Information Gain Formulations
3. Single-Layer Perceptron Learning Rule
4. Multi-Layer Perceptron & Backpropagation
5. Bias-Variance Tradeoff & Regularization

1. Decision Tree Induction: Entropy & Information Gain

$$H(S) = -\sum_{i=1}^{c} p_i \log_2(p_i)$$

$$\mathrm{Gain}(S, A) = H(S) - \sum_{v \in \mathrm{Values}(A)} \frac{|S_v|}{|S|} H(S_v)$$

🏛 **BIT Mesra Exam Question (10 Marks)**

Problem: Explain Backpropagation algorithm for training a Multi-Layer Perceptron with gradient descent derivations.
Solution: Forward pass computes net activations $z_j = \sum w_{ij} x_i$ and outputs $a_j = \sigma(z_j)$. Backward pass propagates error $\delta_k = (y_k - a_k)\sigma'(z_k)$ to hidden layers $\delta_j = \sigma'(z_j) \sum w_{jk}\delta_k$, updating weights via $w_{ij} \leftarrow w_{ij} + \eta \delta_j x_i$.

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